

EXERCISE-2

MANIPULATING DATA

OBJECTIVE

After, the completion of this exercise the students will be able to do the following

- Describe each DML statement
- Insert rows into tables
- Update rows into table
- Delete rows from table
- Control Transactions

A DML statement is executed when you:

- Add new rows to a table
- Modify existing rows
- Removing existing rows

A transaction consists of a collection of DML statements that form a logical unit of work.

To Add a New Row

INSERT Statement

Syntax

INSERT INTO table_name VALUES (column1 values, column2 values, ..., columnn values);

Example:

INSERT INTO department (70, 'Public relations', 100, 1700);

Inserting rows with null values

Implicit Method: (Omit the column)

INSERT INTO department VALUES (30, 'purchasing');

Explicit Method: (Specify NULL keyword)

INSERT INTO department VALUES (100, 'finance', NULL, NULL);

Inserting Special Values

Example:

Using SYSDATE

INSERT INTO employees VALUES (113, 'louis', 'popp', 'lpopp', '5151244567', SYSDATE, 'ac_account', 6900, NULL, 205, 100);

Inserting Specific Date Values

Example:

```
INSERT INTO employees VALUES ( 114,'den', 'raphealy', 'drapheal', '5151274561',  
TO_DATE('feb 3,1999','mon, dd ,yyyy'), 'ac_account', 11000,100,30);
```

To Insert Multiple Rows

& is the placeholder for the variable value

Example:

```
INSERT INTO department VALUES (&dept_id, &dept_name, &location);
```

Copying Rows from another table

➤ Using Subquery

Example:

```
INSERT INTO sales_reps(id, name, salary, commission_pct)  
SELECT employee_id, Last_name, salary, commission_pct  
FROM employees  
WHERE job_id LIKE '%REP');
```

CHANGING DATA IN A TABLE

UPDATE Statement

Syntax 1: (to update specific rows)

```
UPDATE table_name SET column=value WHERE condition;
```

Syntax 2: (To update all rows)

```
UPDATE table_name SET column=value;
```

Updating columns with a subquery

```
UPDATE employees  
SET job_id= (SELECT job_id  
FROM employees  
WHERE employee_id=205)  
WHERE employee_id=114;
```

REMOVING A ROW FROM A TABLE

DELETE STATEMENT

Syntax

```
DELETE FROM table_name WHERE conditions;
```

Example:

```
DELETE FROM department WHERE dept_name='finance';
```

3.

ID	Last_name	First_name	Userid	Salary
1	Patel	Ralph	rpatel	895
2	Dances	Betty	bdances	860
3	Biri	Ben	bbiri	1100
4	Newman	Chad	Cnewman	750
5	Ropebur	Audrey	aropebur	1550

Find the Solution for the following:

1. Create MY_EMPLOYEE table with the following structure

NAME	NULL?	TYPE
ID	Not null	Number(4)
Last_name		Varchar(25)
First_name		Varchar(25)
Userid		Varchar(25)
Salary		Number(9,2)

```
CREATE TABLE employees (
  id NUMBER(4) NOT NULL,
  last_name VARCHAR(25),
  first_name VARCHAR(25),
  userid VARCHAR(25),
  salary NUMBER(9,2),
  CONSTRAINT PK_employees_id PRIMARY KEY (id)
```

2. Add the first and second rows data to MY_EMPLOYEE table from the following sample data.

ID	Last_name	First_name	Userid	salary
1	Patel	Ralph	rpatel	895
2	Dances	Betty	bdances	860
3	Biri	Ben	bbiri	1100
4	Newman	Chad	Cnewman	750
5	Ropebur	Audrey	aropebur	1550

```
Insert All
Into employee (id, last_name, first_name, userid, salary) values (1, 'Patel',
'Ralph', 'rpatel', 895)
Into employee (id, last_name, first_name, userid, salary) values (2, 'Dance',
'Betty', 'bdances', 860)
Display the table with values.
Into employee (id, last_name, first_name, userid, salary) values (3, 'Biri',
'Ben', 'bbiri', 1100)
Into employee (id, last_name, first_name, userid, salary) values (4, 'Newman',
'Chad', 'Cnewman', 750)
SELECT * FROM dual
```

3. Display the table with values?
Select * From employees;

4. Populate the next two rows of data from the sample data. Concatenate the first letter of the first_name with the first seven characters of the last_name to produce Userid.

```
Insert All
Into employees (id, last_name, first_name, userid, salary)
values (6, 'Smith', 'John', (concat(substr('John', 1, 1), substr(
('Smith', 1, 7))), 1200)
Into employees (id, last_name, first_name, userid, salary)
values (7, 'Williams', 'Alice', concat(substr('Alice', 1, 1), substr(
('Williams', 1, 7))), 950)
Select * From dual;
```

5. Make the data additions permanent.

Commit;

6. Change the last name of employee 3 to Drexler.

Update Employees

SET last_name = 'Drexler'

WHERE id = 3;

7. Change the salary to 1000 for all the employees with a salary less than 900.

```
UPDATE Employees  
SET Salary = 1000  
WHERE Salary < 900;
```

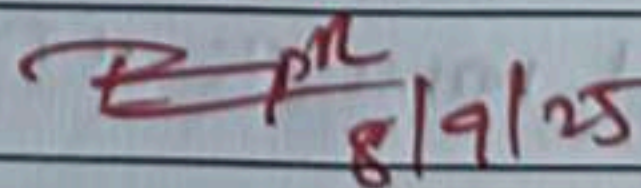
8. Delete Betty dances from MY_EMPLOYEE table.

```
DELETE FROM Employees  
WHERE first_name = 'Betty' AND last_name = 'Dances';
```

9. Empty the fourth row of the emp table.

```
Delete from employees where id = 4;
```



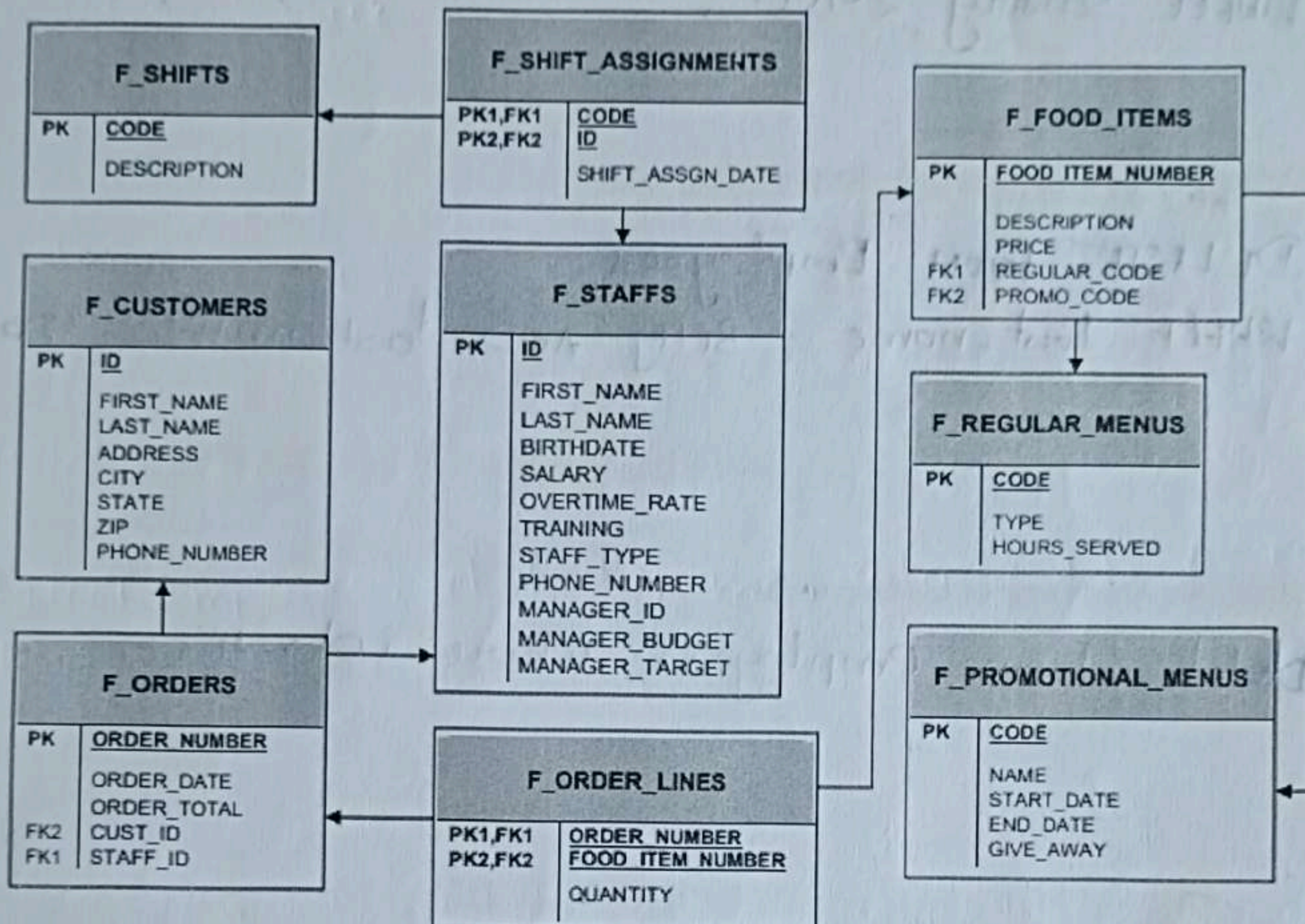
Evaluation Procedure	Marks awarded
Query(5)	5
Execution (5)	5
Viva(5)	5
Total (15)	15
Faculty Signature	 8/9/25

PRACTICE QUESTIONS

Date: _____

Working with Columns, Characters, and Rows

Global Fast Foods Database Tables



1. The manager of Global Fast Foods would like to send out coupons for the upcoming sale. He wants to send one coupon to each household. Create the SELECT statement that returns the customer last name and a mailing address.

Ans: `SELECT last_name, mailing_address FROM Customers;`

2. Each statement below has errors. Correct the errors and execute the query in Oracle Application Express.

a. `SELECT first name FROM f_staffs;` Ans: `SELECT first_name FROM f_staffs;`

b. `SELECT first_name | " " | last_name AS "DJs on Demand Clients" FROM d_clients;` Ans: `SELECT first_name || " " || last_name AS "DJs on Demand Clients" FROM d_clients;`

c. `SELECT DISTINCT f_order_lines FROM quantity;` Ans: `SELECT DISTINCT f_order_lines FROM quantity;`

d. `SELECT order number FROM f_orders;` Ans: `SELECT order_number FROM f_orders;`

3. Sue, Bob, and Monique were the employees of the month. Using the f_staffs table, create a SELECT statement to display the results as shown in the Super Star chart.

Super Star
*** Sue *** Sue ***
*** Bob *** Bob ***
*** Monique *** Monique ***

Ans: SELECT '***' || first_name || '***' || first_name || '***' AS "SUPER STAR"
FROM f_staffs
WHERE first_name IN ('Sue', 'Bob', 'Monique');

4. Which of the following is TRUE about the following query?

SELECT first_name, DISTINCT birthdate
FROM f_staffs;

- a. Only two rows will be returned.
- b. Four rows will be returned.
- c. Only Fred 05-Jan-1988 and Lizzie 10-Nov-1987 will be returned.
- ☒ d. No rows will be returned.

5. Global Fast Foods has decided to give all staff members a 5% raise. Prepare a report that presents the output as shown in the chart.

EMPLOYEE LAST NAME	CURRENT SALARY	SALARY WITH 5% RAISE
Johnson	48000	47250

6. Create a query that will return the structure of the Oracle database EMPLOYEES table. Which columns are marked "nullable"? What does this mean?

DESC employees;
if a column is marked as NULL, it means the column can accept null values, indicating that data is optional for that column.

7. The owners of DJs on Demand would like a report of all items in their D_CDs table with the following column headings: Inventory Item, CD Title, Music Producer, and Year Purchased. Prepare this report.

SELECT inventory_item, cd_title, music_producer, year_purchased
FROM d_cds;

8. True/False - The following SELECT statement executes successfully: SELECT last_name, job_id, salary AS Sal FROM employees;

SELECT last_name, job_id, salary AS Sal FROM employees; True

9. True/False - The following SELECT statement executes successfully: SELECT * FROM job_grades;

True

10. There are four coding errors in this statement. Can you identify them?

SELECT employee_id, last_name sal x 12 ANNUAL SALARY FROM employees;

SELECT employee_id, last_name, salary * 12 AS annual_salary FROM employees;

11. In the arithmetic expression salary * 12 - 400, which operation will be evaluated first?

*

12. Which of the following can be used in the SELECT statement to return all columns of data in the Global Fast Foods f_staffs table?

a. column names

☒ b. *

c. DISTINCT id

d. both a and b

13. Using SQL to choose the columns in a table uses which capability?

a. selection

☒ b. projection

c. partitioning

d. join

14. SELECT last_name AS "Employee". The column heading in the query result will appear as:

a. EMPLOYEE

☒ b. employee

☒ Employee

d. "Employee:

15. Which expression below will produce the largest value?

a. `SELECT salary * 6 + 100`

☒ b. `SELECT salary * (6 + 100)`

c. `SELECT 6(salary + 100)`

d. `SELECT salary + 6 * 100`

16. Which statement below will return a list of employees in the following format?

Mr./Ms. Steven King is an employee of our company.

a. `SELECT "Mr./Ms." || first_name || ' ' || last_name 'is an employee of our company.' AS "Employees"`
`FROM employees;`

b. `SELECT 'Mr./Ms. ' || first_name || last_name || ' ' || 'is an employee of our company.' FROM`
`employees;`

☒ c. `SELECT 'Mr./Ms. ' || first_name || ' ' || last_name || ' ' || 'is an employee of our company.' AS`
`"Employees" FROM employees ;`

d. `SELECT Mr./Ms. || first_name || ' ' || last_name || ' ' || "is an employee of our company." AS`
`"Employees" FROM employees`

17. Which is true about SQL statements?

a. SQL statements are case-sensitive

b. SQL clauses should not be written on separate lines.

☒ c. Keywords cannot be abbreviated or split across lines.

d. SQL keywords are typically entered in lowercase; all other words in uppercase.

18. Which queries will return three columns each with UPPERCASE column headings?

a. `SELECT "Department_id", "Last_name", "First_name"`
`FROM employees;`

☒ b. `SELECT DEPARTMENT_ID, LAST_NAME, FIRST_NAME`
`FROM employees;`

c. `SELECT department_id, last_name, first_name AS UPPER CASE`
`FROM employees`

d. `SELECT department_id, last_name, first_name`
`FROM employees;`

19. Which statement below will likely fail?

☒ a. `SELECT * FROM employees;`

b. `Select * FROM employees;`

c. `SELECT * FROM EMPLOYEES;`

d. `SelecT* FROM employees;`

20. Click on the History link at the bottom of the SQL Commands window. Scroll or use the arrows at the bottom of the page to find the statement you wrote to solve problem 3 above. (The one with the column heading SuperStar). Click on the statement to load it back into the command window. Execute the command again, just to make sure it is the correct one that works. Once you know it works, click on the SAVE button in the top right corner of the SQL Commands window, and enter a name for your saved statement. Use your own initials and "_superstar.sql", so if your initials are CT then the filename will be CT_superstar.sql.

Log out of OAE, and log in again immediately. Navigate back to the SQL Commands window, click the Saved SQL link at the bottom of the page and load your saved SQL statement into the Edit window. This is done by clicking on the script name. Edit the statement, to make it display + instead of *. Run your amended statement and save it as initials_superplus.sql.

This is more of an action you need to take in Oracle Application Express (OAE) it involves saving your query for problem 3, logging out, and then logging back in to re-execute the query

Evaluation Procedure	Marks awarded
Practice Evaluation (5)	5
Viva(5)	5
Total (10)	10
Faculty Signature	<i>[Signature]</i> 6/9/25